

University of Westminster
University Research Ethics Committee

Application for Research Ethics

PART A

Section 1 – PROJECT AND APPLICANT DETAILS

1.1 Project Title: Adaptive Machine Learning System for Personalized Mental Health Indicators Detection in Social Media Behavior

1.2 Applicant Details

Name: A.M.K.P.Abeysundara **University Email Address:** 20210410@iit.ac.lk

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Faculty: School of Computing

Please check the relevant box:

Undergraduate ☒ Postgraduate ☐ MPhil/PhD Student ☐ Staff ☐

I confirm I have read the *University's Code of Practice Governing the Ethical Conduct of Research* YES ☒ NO ☐

1.3 Supervisor/Dean of Faculty/Faculty Research Director details

Please note that all applicants with a supervisor(s) must ensure that the supervisor signs the declaration at the bottom of this page if completing Part A only or in **Section 10.3** if completing Part B

All **staff** must ensure that their Dean of Faculty, or Faculty Research Director (or nominee), as appropriate, signs the declaration at the bottom of this page if completing Part A only or in **Section 10.3** if completing Part B

Name: Prasandika Balage **University Email Address:** prasandika.b@iit.ac.lk

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Prasandika Balage
Digitally Signed on 11/11/2025

PART A (Continued)

Section 2 – Project Details

2.1 Please provide a description of the background with references to relevant literature (250 words maximum): Mental health is a condition that occurs in almost one out of every eight people throughout the world and is one of the most prevalent causes of disability and a low-quality of life. Even though there have been innovations in the field of psychiatry, it is still challenging to diagnose schizophrenia in its early stages, with nearly half of the victims going undiagnosed because of insufficient access to professional screening (WHO, 2022). The social media network including Twitter has more than 5 billion users, and is now colossal stores of behavioural and linguistic information that capture the psychological conditions, moods, and interactions. Recent research allows showing that early signs of depression, anxiety, and stress can be detected through digital behavioural patterns (Chancellor and De Choudhury, 2020; Mansoor and Ansari, 2024).

In computational psychiatry and digital phenotyping, machine learning has started to be used to discover these patterns but most current systems are not intended to be used to tell an individual but instead to monitor them within an institution (Benrimoh et al., 2023). This establishes a huge gap in research linking clinical level of detection and user level of mental health awareness. Integrating both natural language processing (BERT) and time modeling (LSTM) as proposed in this study will help turn individual social media activity into explainable variables of mental health change. The project adds to the accumulated experience in the area of personalized mental health analytics, with a background of consideration of ethical data use as well as preservation of privacy, and encouragement of early, non-clinical self-awareness over diagnosis.

2.2. Please provide a brief description and the aims of your study (250 words maximum): The paper, MoodMirror: Adaptive Machine Learning System of Personalized Mental Health Indicators Detection in Social Media Behaviour, is a proposal to develop and test a machine learning system that would be able to detect early indicators of mental health in both linguistic and behavioural indicators in social media usage.

The system will be used to identify the relationship between language patterns, posting habits, and interaction patterns and emotional well-being with the help of the publicly available Twitter and Reddit data that reveals textual, temporal, and engagement-based features. The given hybrid BERT + LSTM model combines the view of the contextual understanding of language with time-related behaviour analysis to identify the anomalies in the normal behaviour of a person online.

The objectives are to:

Build an individualized privacy-sensitive mental health awareness analytic model.

Assess the model accuracy, interpretability and ethical safety of generating non-diagnostic feedback.

This is purely academic research. The official Twitter Academic API and Reddit API will be used to gather all the information in a manner that is ethically acceptable and only publicly available data will be used. Anonymous or anonymization of personal identifiable information will be done. The final vision aims at advocating proactive mental health self-awareness in a responsible transparent and user-friendly computational approach.

2.3. Please outline the design and methodology of your study (include details of the selection and recruitment of participants (if any) and details of any invasive (e.g. blood samples, inhalation/ingestion of food and/or non-food products (in abnormally higher or lower levels than normal or a different form), or intrusive (e.g. questionnaires, focus groups, interviews, etc.) procedures [attach extra information as necessary] (400 words maximum in total): According to this study, the non-experimental, quantitative design is based on the secondary data gathered in an ethical manner using the two open social media, including Reddit and X

(Twitter). Access to data will be done via their official public APIs, the Python Reddit API Wrapper (PRAW), and the Twitter Academic Research API to ensure the policies of the platforms and ethical principles to be followed. There will be no human interaction, recruitment or intervention.

Data Source and Selection:

The posts and comments will be obtained on the publicly available communities on Reddit (e.g., r/depression, r/Anxiety, r/mentalhealth) and similar Twitter debates where individuals willingly share their experiences of mental health issues. The data to be collected will be confined to English-language publicly available text data. Personally identifiable information (usernames, links, or geolocation), private accounts, direct message or deleted content will be excluded. The records will be anonymized on the spot by hashing the remaining identifiers.

Ethical Safeguards:

The ethical guidelines of the research followed in the study include the University of Westminster and Informatics Institute of Technology, as well as GDPR rules of data minimization and privacy. Because the data is publicly available and business is not required to employ any consent mechanism, the posts will be handled in a sensitive and respectful way in order to prevent stigmatization and harm. The ethical consequence of the project is very low because it does not include any human participants or subjects of the invasive procedures and does not imply direct contact with users.

Analysis and Data Processing:

Textual information will be subjected to cleaning and normalization through the elimination of noise (e.g. URLs, emojis) and the preservation of valuable emotional and linguistic information. The hybrid BERT + LSTM model will be used to analyze the semantic and temporal features:

BERT learns contextual meaning and tones.

LSTM learns the behavioral patterns like posting frequency and engagement patterns.

The 30-day posting record of each anonymized user will initiate a personal baseline based on z-score normalization to identify the deviation that can potentially be an early sign of psychological distress. The evaluation of model accuracy will be based on the measures of precision, recall, F1-score and AUC-ROC.

Data Security:

Each of the datasets will be stored in secured and encrypted access-controlled space (institutional servers and restricted Google Colab sessions). It will be accessible only to the principal researcher and supervisor. No information would be distributed or sold to the outside world.

To sum up, the project is a privacy-preserving computational framework on a publicly available Reddit and Twitter dataset meant to create a machine learning-based system of early mental health awareness without violating user privacy.

2.4. Timescales

Start Date (DD/MM/YY): 01/09/2025

Estimated duration of work: 8 -10 months

Section 3 - RISK OF HARM				
NOTE 1: Where indicated below applicants should check if the research will require ethical approval from a National Research Ethics Committee via the Integrated Research Application System (IRAS) - nres.queries@nhs.net - http://www.hra-decisiontools.org.uk/ethics/ NOTE 2: The University of Westminster holds a Human Tissue Authority Licence – This licence is specifically for tissue stored at 115 New Cavendish Street in accordance with the terms of the licence – Advice must be obtained from the University Human Tissue Designated Individual (N.Presneau@westminster.ac.uk)				
RISK OF HARM (to self, colleagues, participants, environment or animals)		Yes	No	N/A
1	Will any pain or more than mild discomfort result from the study?	☐	☒	☐
2	Could the study induce any psychological stress or anxiety or cause harm or negative consequences beyond the risks encountered in normal life?	☐	☒	☐
3	Will the study involve prolonged or repetitive physical or psychological testing of human participants that may put someone at risk, e.g. use of treadmill?	☐	☒	☐
4	Will the study involve raising sensitive topics (e.g. sexual activity, drug use, revelation of medical history, bereavement, illegal activities, etc.)?	☐	☒	☐
5	Does your work involve any "relevant material" containing human cells (e.g. blood, urine, saliva, body tissues but NOT established cell-lines) from living or deceased persons (Such work must take account of the Human Tissue Act)? – See Note 1 and 2 above.	☐	☒	☐
6	Will DNA samples be taken from human participants (Such work must take account of the Human Tissue Act)? – See Note 1 and 2 above.	☐	☒	☐
7	Does your study raise any issues of personal safety for you or other researchers or participants involved in the project (Especially relevant if taking place outside working hours or off University premises)?	☐	☒	☐
8	Does your study involve deliberately misleading the participants (e.g. deception, covert observation)?	☐	☒	☐
9	Does your work involve administration of a food or non-food substance of a different type from or in abnormally higher or lower amounts than normal or one that is known to cause allergic reaction(s) or potential psychological stress?	☐	☒	☐
10	Does your study involve issues relating to personal and/or sensitive data?	☐	☒	☐
PARTICIPANTS (and/or their records/associated data)		Yes	No	N/A
Does your work involve any of the following:				
11	Human participants in a health and/or social care setting (e.g. patients, those attending day centres, community care, rehabilitation centres, etc., including in the NHS, other public, private and/or voluntary sectors)? – See Note 1 above.	☐	☒	☐
12	Human participants who may be deemed vulnerable (e.g. children, people in poverty and/or with physiological or psychological impairments, persons attending rehabilitation centres, persons in easily identifiable positions that could be subject to victimisation, etc.)?	☐	☒	☐
13	Expectant or new mothers?	☐	☒	☐
14	Refugees/Asylum seekers?	☐	☒	☐
15	Minors (under the age of 18 years old)?	☐	☒	☐
16	Participants in custody (e.g. prisoners or arrestees)? – See Note 1 above.	☐	☒	☐
17	Participants with impaired mental capacity (e.g. severe mental illness, brain damage, sectioned under Mental Health Act, lowered or reduced sense of consciousness)? – See Note 1 above.	☐	☒	☐
18	Animals (or animal tissue).	☐	☒	☐
INFORMATION TO PARTICIPANTS		Yes	No	N/A
19	Will you provide participants with a Participant Information Sheet prior to obtaining informed consent which can be taken away by the participant?	☐	☐	☒
20	Will you describe the procedures to participants in advance, so that they are informed about what to expect?	☐	☐	☒
21	Will you obtain informed consent for participation (normally written)? OR in the case of using personal data previously acquired was consent given for the reuse of the data for other research purposes?	☐	☐	☒
22	Will you tell participants that they may withdraw from the research at any time and for any reason without any impact on their care, service provision etc.?	☐	☐	☒
23	Will you give participants the option of omitting questions they do not want to answer?	☐	☐	☒
24	Will you tell participants that their data will be treated as confidential and that, if published, it will not be identifiable as theirs?	☐	☐	☒
25	Will you offer feedback to participants at the end of their participation, upon request (e.g. give them a brief explanation of the study and its outcomes)?	☐	☐	☒
26	Has external funding or collaboration been applied for/received, which requires institutional ethical consideration or approval?	☐	☐	☒

Useful links:

- <http://www.screc.org.uk/> - Social Care Research Ethics Committee
- <http://www.hra-decisiontools.org.uk/ethics/> - Human Research Authority decision tool to identify if research needs National Research Ethics Committee approval
- <http://www.nres.nhs.uk/applications/guidance/governance-and-directives/?entryid62=131341> – Governance Arrangements for Research Ethics Committees
- <http://www.nres.nhs.uk/EasySiteWeb/GatewayLink.aspx?allId=134016> - NRES algorithm “Does my project require review by a Research Ethics Committee”?
- <http://www.hta.gov.uk/policiesandcodesofpractice/codesofpractice.cfm> - Human Tissue Authority Code of Practice
- <http://www.hta.gov.uk> – Human Tissue Authority website
- http://www.rsclern.mrc.ac.uk/MRC_HumanTissueAct/player.html - Medical Research Council online training course for Human Tissue Act.

What to do next:

- **If you have answered NO to questions 1-18 (inclusive) and YES to questions 19-25 (inclusive)**, you do not need to complete the Full Research Ethics Approval Form (Part B). Please keep this form for your records, and do not submit to **Faculty Research Ethics Committee (FREC)** unless you require ethical consideration of your study, regardless of ethical implications, by an external body (question 26 has been answered YES). **A list of Faculty contacts is below.**
- **If you have answered YES to any of the questions 1-18 (inclusive) or NO to any of the questions 19-25** the Full Research Ethics Approval Form (Part B) **MUST** be submitted including Cover Sheet, Part A and Part B of the application form plus any required supplementary documents to the Secretary of the relevant Faculty Research Ethics Committee (FREC). **A list of Faculty contacts is below.**
- If you are applying for external Ethical Approval, please send a *copy* of the Conditions/Approvals letters to the **University Research Ethics Committee (UREC)** Secretary (this may include the original ethical application(s)). Where the external ethics committee/body has equal standing or primary jurisdiction, e.g. another University Research Ethics Committee or a National Research Ethics Committee, any approval will normally be received and noted by the University of Westminster Research Ethics Committee and further consideration may not be required. Where the external committee does not have equal or higher standing than the University Committee then the full ethical approval process at the university may still be required. Additional institutional compliance issues may need consideration by UREC.
- All Applications (dated, signed and authorised) and supplementary information or External Approvals should be sent to the University Research Ethics Committee (UREC) Secretary *in electronic format with a version number, document name and date and the Principal Investigator (or Undergraduate/Postgraduate Taught Student) name. On receipt your application will be issued a unique reference number*
- All new Applications should be submitted to a Research Ethics Committee (FREC or UREC) Secretary a minimum of 10 working days in advance of the Committee meeting date (earlier submission is recommended so that applications can be pre-vetted and obvious issues addressed before the application is considered by the Committee).

Contact details:

<u>Faculty</u>	<u>Chair</u>	<u>Secretary</u>
<u>Architecture and the Built Environment</u>	<u>Professor Nick Bailey</u>	<u>Colette Davis</u>
<u>Media Arts and Design</u>	<u>Dr Anthony McNicholas</u>	<u>Fauzia Ahmad</u>
<u>Science and Technology</u>	<u>Dr John Colwell</u>	<u>Mandy Walton</u>
<u>Science and Technology Psychology Department Sub Committee</u>	<u>Dr Laura Boubert</u>	<u>TBC</u>

Faculty	Chair	Secretary
Social Sciences and Humanity	Professor Marco Roscini	Victoria Grey-Edwards
Westminster Business School	Petar Sudar	Haydn Worley
University Research Ethics Committee	Professor Graham Megson	Huzma Kelly

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